



Project 4 - ALY6000 R Independent Data Analysis

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ALY 6000 Introduction to Analytics

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Dataset Overview

This analysis utilizes the IBM HR Analytics Employee Attrition & Performance Dataset obtained from [Kaggle](#), containing 1,470 employee records across 35 original variables. The dataset includes diverse variable types:

Categorical (Nominal): Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, OverTime, Over18

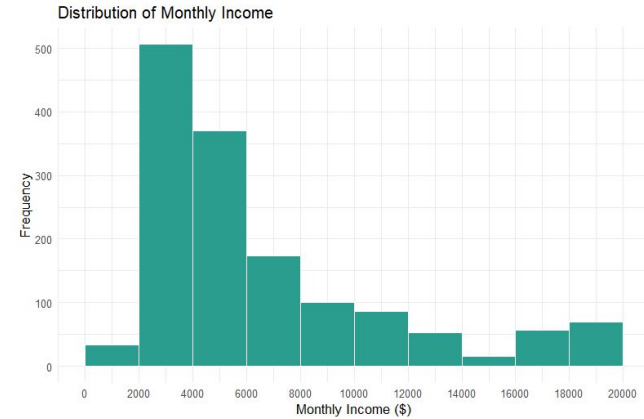
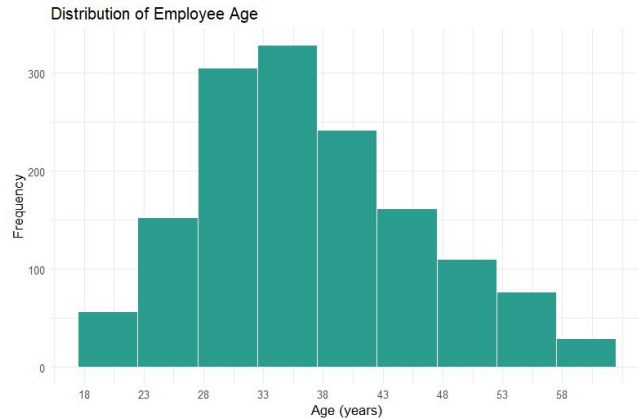
Categorical (Ordinal): Education (1-5), EnvironmentSatisfaction (1-4), JobInvolvement (1-4), JobLevel (1-5), JobSatisfaction (1-4), PerformanceRating (1-4), RelationshipSatisfaction (1-4), StockOptionLevel (0-3), WorkLifeBalance (1-4)

Numerical (Ratio): Age, DailyRate, DistanceFromHome, HourlyRate, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, TotalWorkingYears, TrainingTimesLastYear, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager

Identifier: EmployeeNumber

Constant: EmployeeCount (all=1), StandardHours (all=80)

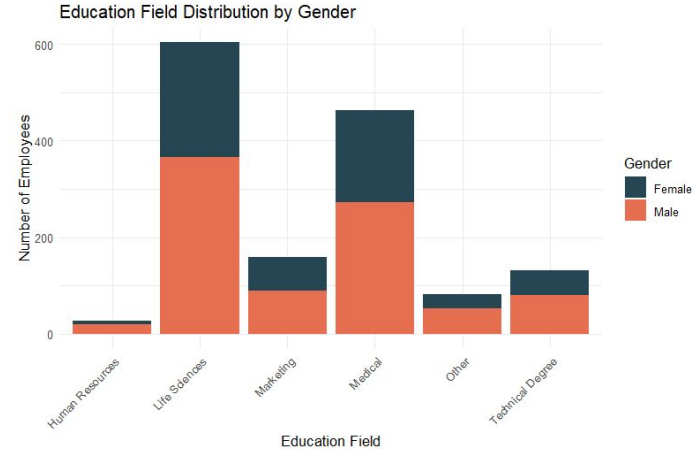
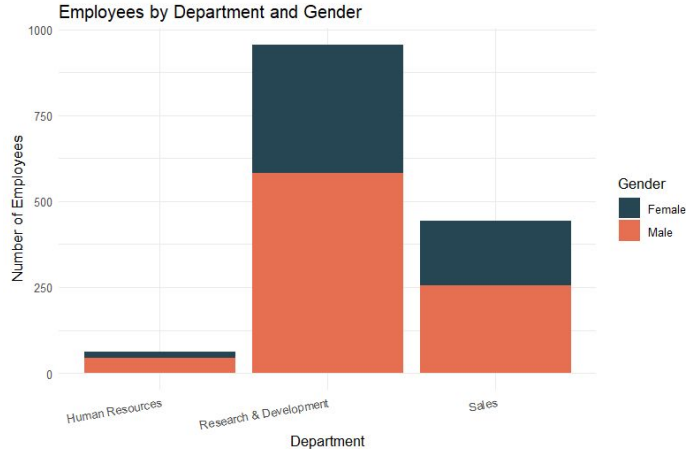
Employee Age and Monthly Income Distribution



Age Distribution: Employee age varies between 18-60 years, with average of 36.9 years. Bell-shaped distribution with peak at 33-38 years, followed by 28-33 years. Age 30-43 range contains 39% of workforce. Declining trend after age 43. Only 4% employees below age 23. Balanced mix of youth and experience.

Income Distribution: Monthly income varies between 1,000-20,000, with average of 6,510. Right-skewed distribution with 60% earning between 2,000-6,000. Only 13% earn above 10,000. Largest concentration at lower-middle income range (2,000-6,000). Shows typical pyramid structure.

Department and Education Field with Gender

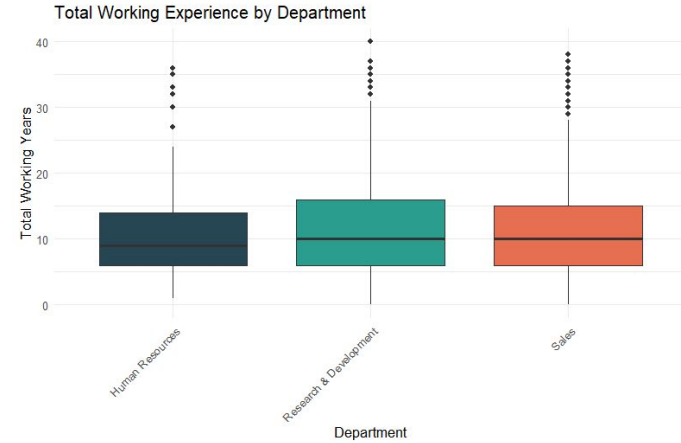
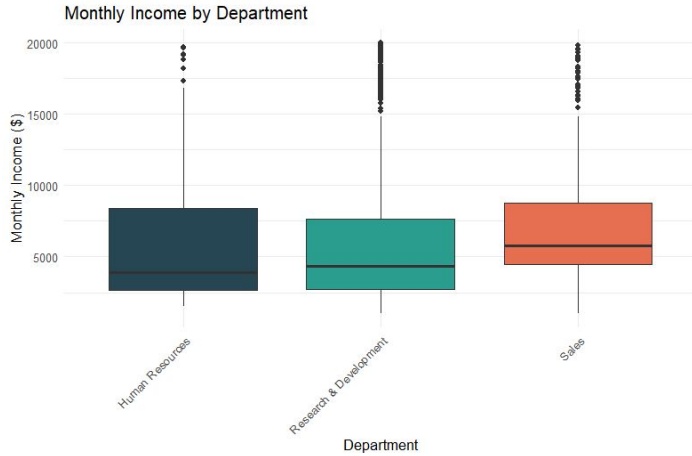


Department & Gender Distribution: R&D dominates the organisation with 960 employees (65%), followed by Sales (440, 30%) and HR (70, 5%). This structure reflects strong innovation and product development focus.

Gender distribution shows 60-40 male-female ratio overall, with R&D maintaining similar balance. Sales has more male (65-35), while HR achieves near parity.

Education Field & Gender Distribution: Life Sciences (600+, 41%) and Medical backgrounds (460, 31%) dominate, together comprising 72% of workforce. This scientific concentration aligns with R&D department size and suggests pharmaceutical or biotech industry. Marketing (160) shows balanced gender representation, while Technical Degrees (140) remain male-concentrated. HR specialization is smallest at just 27 employees (2%).

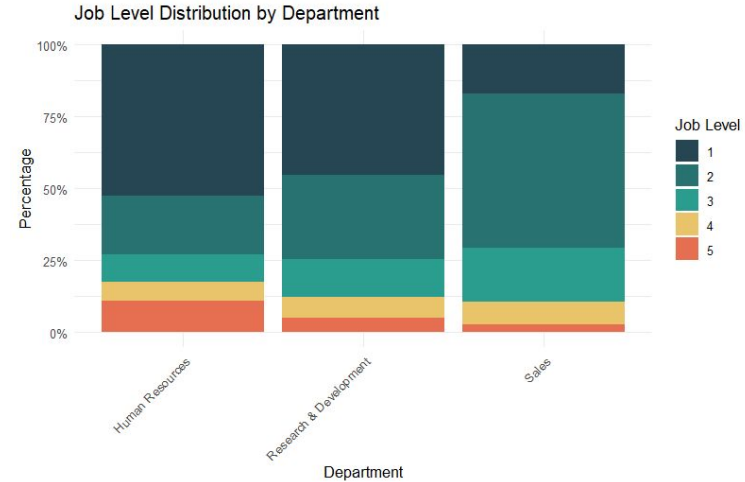
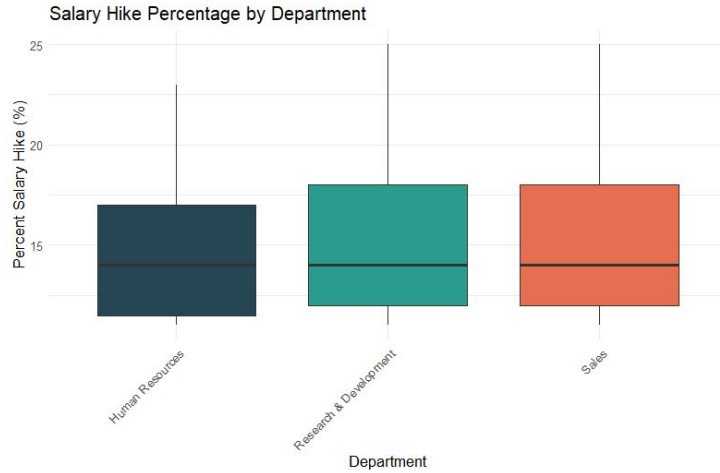
Department wise Monthly Income and Experience



Monthly Income by Department: Sales shows highest median income at 5,754, followed by R&D at 4,374 and HR at 3,886. Sales has tightest interquartile range (4,459-8,800) concentrated at upper-middle income, while HR (2,644-8,412) and R&D (2,741-7,655) show wider spreads starting from lower levels. Sales exhibits extreme outliers reaching 20,000 compared to other departments' maximum of 17,000-18,000, suggesting commission-based or performance-driven compensation creating high-earner opportunities but also less income instability.

Total Working Experience by Department: All departments show nearly identical median experience around 10 years with similar interquartile ranges. Multiple outliers exist at 35-40 years across all functions. Despite similar experience levels across departments, income varies significantly - suggesting that department choice and job role impact salary more than years of experience.

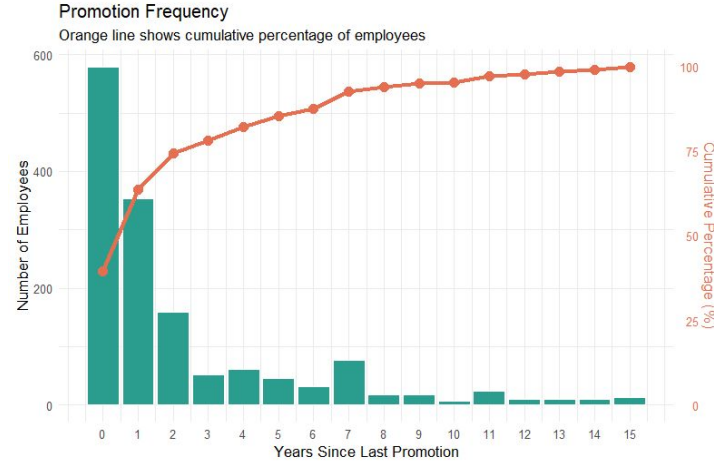
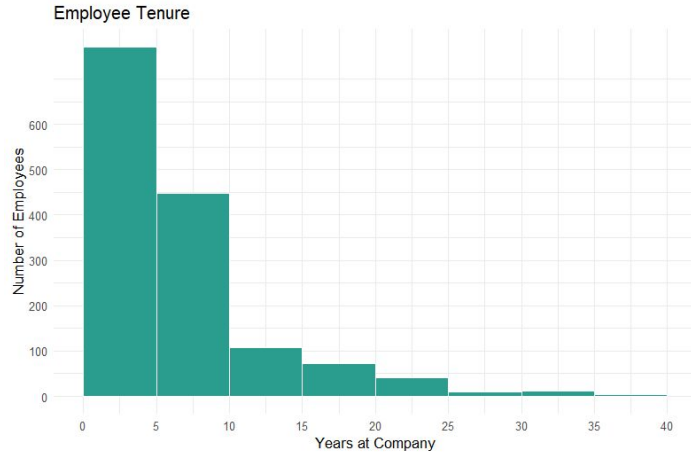
Department wise Salary hike and Job level



Salary Hike Percentage by Department: All three departments show same median salary hike at 14%, with similar interquartile ranges: HR (11.5%-17%), R&D (12%-18%), and Sales (12%-18%). However,- R&D and Sales have outliers reaching up to 25%, while HR's maximum appears around 23%. This suggests that while the company maintains consistent median hikes across departments, R&D and Sales employees have better chances of receiving exceptional raises (up to 25%), possibly reflecting higher performance variability or promotion opportunities.

Job Level Distribution by Department: HR is heavily loaded at Level 1 (52%) with very few people at higher levels - only 11% reach Level 5. This means limited growth opportunities in HR. R&D shows better balance with 45% at Level 1, steadily decreasing to 5% at Level 5, showing clear career path. Sales has unusual pattern - with 54% at Level 2, meaning most sales jobs need experience. Sales has very few senior positions (3% Level 5), which may show experienced salespeople leave for better growth opportunities elsewhere.

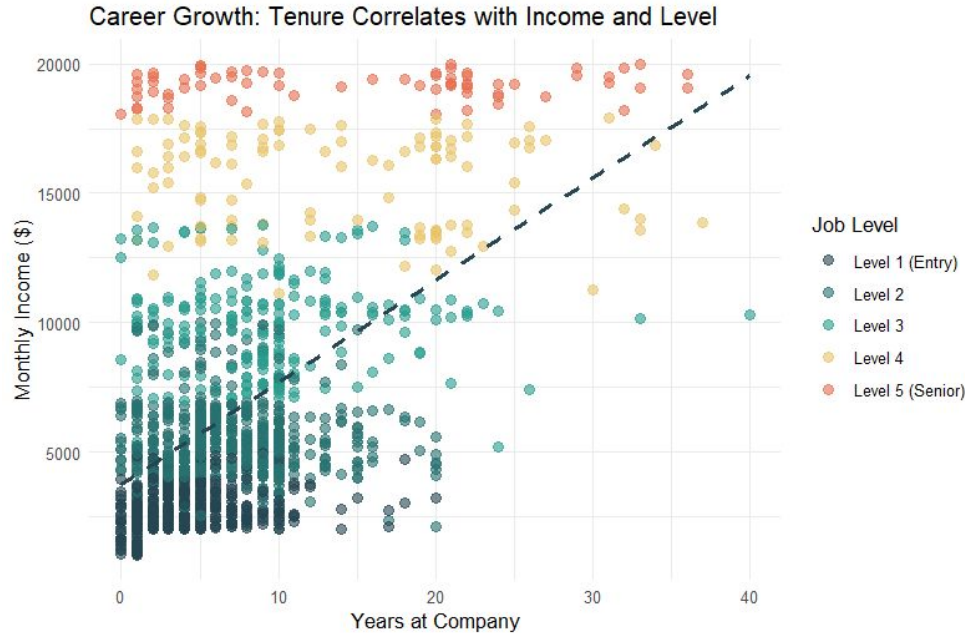
Employees Tenure and Last Promotion



Employee Tenure Distribution: Tenure shows sharp right-skewed distribution with highest concentration at 0-5 years over 54%. The distribution drops dramatically to 430 employees so on, and beyond 20 years very few remain. This steep drop after 5 years indicates either employees are leaving the company or the organization grew significantly in recent years by hiring many new people. Less than 15% of employees stay beyond 10 years, which may create problems for retaining experienced staff and developing future leaders.

Promotion Frequency: Promotion patterns show interesting trends. Around 580 employees (39%) got promoted in current year (0), and 350 employees (24%) got promoted 1 year ago. Together, 63% of employees received promotion within last year. The orange cumulative line rises sharply, reaching 75% by year 2 - meaning three out of four employees got promoted within last 2 years. After year 2, the curve flattens, showing slower promotion activity. Small groups of employees are spread between years 3-7 without promotion (30-70 employees each year), with an unusual spike at year 7 (around 70 employees). Some employees have waited 10-15 years for promotion, though these are few. This pattern suggests most employees get promoted quickly in first 2 years, but after that, promotion chances reduce significantly, creating career stagnation for those who miss early promotions. Showing characteristics of ogive chart.

Career Growth: Tenure vs Income by Job Level



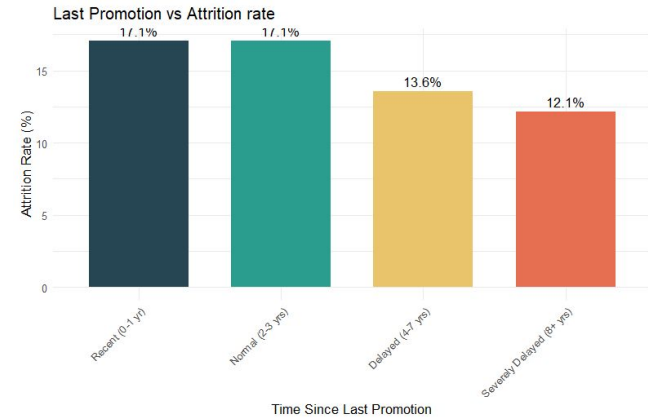
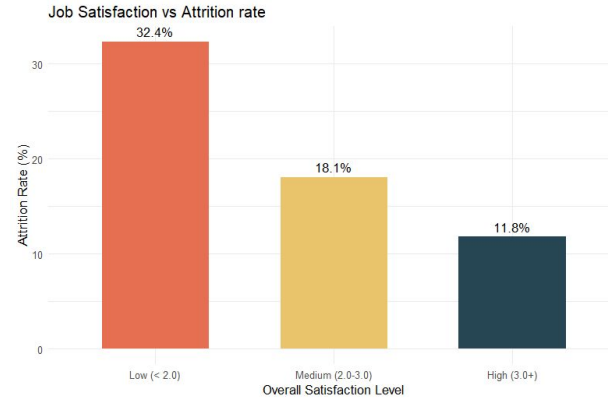
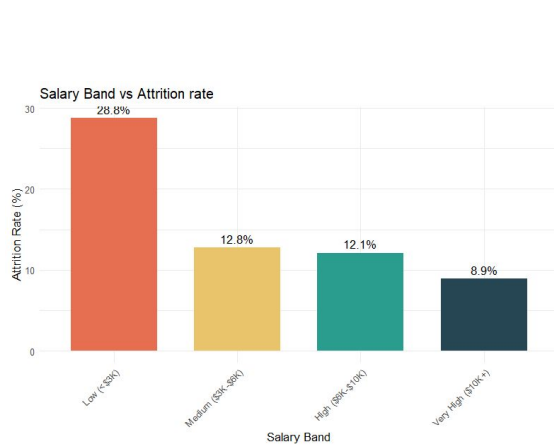
The scatter plot shows income increases with both tenure and job level, but job level is more important.

Level 1 (entry) employees cluster at bottom with 2,000-5,000 income, mostly having 0-10 years tenure. Level 2 spreads between 5,000-10,000, Level 3 at 8,000-13,000, Level 4 at 12,000-17,000, and Level 5 (senior) at 17,000-20,000.

The trend line shows positive relationship between tenure and income. However, dense vertical clustering at 0-5 years across all levels indicates the company hired at all levels recently, not just entry positions.

Some employees with 20-30 years tenure remain stuck at Levels 1-2 earning under 5,000, showing career stagnation where long tenure did not lead to promotion. Wide income spread at same tenure points confirms advancement to higher job levels matters more than just staying longer in the company.

Attrition: Salary, Job Satisfaction, Last Promotion



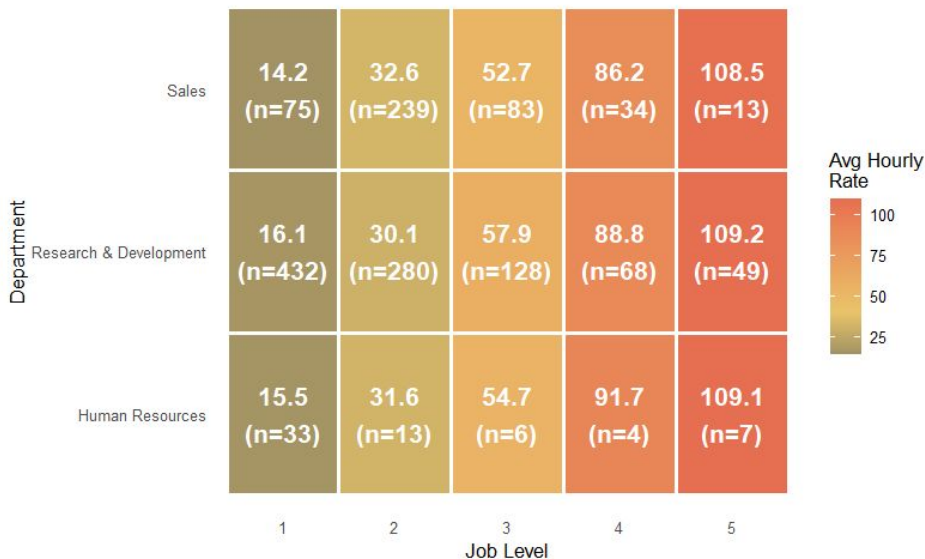
Salary Band vs Attrition: Low salary band shows highest attrition at 28.8% and decreases continuously. Clear pattern: as salary increases, attrition decreases. The sharp drop from 28.8% to 12.8% indicates under 3k is critical threshold where employees actively seek better pay elsewhere. Beyond 6k salary becomes less important as other factors take over.

Job Satisfaction vs Attrition: Low satisfaction shows 32.4% attrition - highest among all factors. Medium drops to 18.1%, and high reduces to 11.8%. Low satisfaction employees are three times more likely to leave than highly satisfied ones. Satisfaction has stronger impact than salary making it the most powerful attrition predictor.

Last Promotion vs Attrition: Recent promotion shows 17.1% attrition, same as normal cycle at 17.1%. Delayed promotions show lower 13.6%, and severely delayed shows 12.1%. This unexpected pattern suggests recent promotions may have been "too late" for already-disengaged employees, or promoted employees become more marketable and leave anyway. Long-delayed employees may have accepted their situation.

Hourly Rate with Department and Job level

Hourly Rate: Department × Job Level



The heatmap shows clear hourly rate progression from Level 1 (14-16) to Level 5 (108-109) across all departments, representing approximately 7x increase.

Interestingly, all departments pay nearly identical rates at same job levels, indicating company-wide standardized compensation policy rather than department-specific pay.

For example, Level 5 converges at around 109 per hour across all departments. R&D pays slightly more at entry levels (16.1 vs 14.2 in Sales), possibly to attract technical talent.

The color gradient from brown to orange visually confirms compensation is determined by job level, not department choice. Employee distribution reveals structural differences: R&D shows healthy pyramid (432 at Level 1, 49 at Level 5), while Sales concentrates at Level 2 (239 of 444 employees) and HR at Level 1 (33 of 63 employees), limiting advancement opportunities in these departments.

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